NATIONAL
Junior College

SH2 Preliminary Examination 2025

9173 H2 Geography Paper 2

Suggested Marking Guide

Disclaimer:
The Suggested Marking Guide should be treated with confidentiality, and should be used with discretion.

Notes on Marking

Marking Approaches

a. **Point marking** is used for marking all sub-parts of the structured questions, except for the 10-mark fieldwork evaluative sub-part.

   For point marking, the general rule is that each creditworthy response is awarded one mark. More detailed guidance is provided in the marking guide for each question.

b. **Generic level descriptors** are used for marking:

- 10-mark fieldwork evaluative sub-part structured question; and
- 20-mark essays.

Possible approaches are provided for questions marked using generic level descriptors. They provide guidance on how the questions could be approached. The possible approaches provided are **neither exhaustive nor should they be treated as model approach to questions**.

Generic Level Descriptors

Levels	Marks	Generic Level Descriptors for H2 Fieldwork Evaluative Question
3	9–10	Evaluation is analytical and coherent. Response addresses the question and demonstrates good knowledge and understanding of fieldwork methods relevant to the given context.
2	5–8	Evaluation is mostly analytical and coherent. Weaker responses in this level will have evaluation that is broadly analytical and generally coherent. Response generally addresses the question and demonstrates adequate knowledge and understanding of fieldwork methods relevant to the given context.
1	1–4	Response is descriptive with limited or no analysis and evaluation. Response is fragmented and lacks clarity. Response lacks focus on the question and may be largely irrelevant to it. Response shows basic knowledge and understanding of fieldwork methods. Response has some, though limited, relevance to the given context.
0	0	No creditworthy response

Levels	Marks	Generic Level Descriptors for H2 Essays
5	18–20	Evaluation is consistently analytical and coherent. Response is well-supported by relevant material, including the effective use of examples. Response features accurate geographical knowledge and reflects good understanding of the subject content relevant to the question.
4	14–17	Evaluation is analytical and coherent. Response is mostly well-supported by relevant material, including the appropriate use of examples. Response features accurate geographical knowledge and reflects adequate understanding of the subject content relevant to the question.
3	10–13	Evaluation is broadly analytical and generally coherent. Response is moderately well-supported by relevant material, including some appropriate use of examples. Response features accurate geographical knowledge and reflects adequate understanding of the subject content relevant to the question.
2	6–9	Response is largely descriptive with limited analysis and evaluation. Response is partly coherent and may lack clarity in parts. Response is poorly supported by relevant materials, including the limited use of examples. Response features inaccurate geographical knowledge and poor understanding of the subject content relevant to the question.
1	1–5	Response is descriptive with no analysis or evaluation. Response is fragmented and lacks clarity. Response consists of unsupported assertions. Response features largely inaccurate geographical knowledge and a lack of understanding of the subject content relevant to the question.
0	0	No creditworthy response

Note:
Assessment using generic level descriptors involves qualitative rather than quantitative evaluation. Judgements on the level to be awarded to an answer will be based on the principle of 'best fit' determined by the descriptors within each level.

As a general guideline, responses deemed to have fulfilled all the descriptors within a level may be awarded the top mark in that level.

Section A

Cluster 4: Fieldwork

Question	Answer	Marks
	A group of six students in Singapore carried out fieldwork to investigate how to engage youth in recycling efforts and encourage them to adopt environmentally sustainable practices through textile recycling and upcycling. The hypothesis for their investigation was: 'Youth aged between 15 and 35 years old residing in Serangoon are more likely to participate in textile recycling and upcycling if they have greater awareness of the environmental impact of the textiles industry and textile waste.' To test their hypothesis, they decided to observe people who deposited their clothes into the textile recycling bins at three chosen sites in an urban neighbourhood, Serangoon, on a Friday during the June holidays. The students split themselves into three groups, and collected data at the three sites concurrently. The neighbourhood was chosen due to its proximity to their school. At each site, the students sat nearby to record the total number of people who deposited their clothes and noted the number of youth after asking them if they were aged between 15 and 35. By using convenience sampling, they administered a survey to 30 respondents aged between 15 and 35 at the three sites, with 10 respondents per site, to better understand their level of awareness of the environmental impact of the textiles industry and textile waste, as well as their level of participation in textile recycling and upcycling. They obtained consent and veified their ages before starting the survey. At the same time, an in-depth interview was conducted with six of the respondents to gain deeper insights. Based on their data collected, they concluded their hypothesis was valid.	

Question	Answer	Marks
1(a)	**Resource 1A is a map showing the locations of the three chosen sites in Serangoon marked as X, Y and Z. Resource 1B shows photographs of the three chosen sites. Resource 2 shows the data collected at each site. Resource 3 shows the questionnaire survey used by the students. Resource 4 shows selected results of the questionnaire survey. Resource 5 shows selected excerpts of the in-depth interviews.** **Explain why this hypothesis is appropriate for the fieldwork.** *Award 1 mark for each explanation of why the hypothesis is appropriate, to a maximum of 4 marks.* *Award a maximum of 1 additional mark for further development of each explanation, where applicable.* Possible responses include: • The hypothesis is **clearly defined** as it identified the **specific target group** which are youth aged between 15 and 35 for the fieldwork. • It also clearly defines the **focus of the fieldwork**, i.e. youth's awareness of environmental impact of textiles industry and textiles waste and their efforts in recycling / upcycling of textiles to be investigated. • It is at a **suitable scale** as the investigation is confined in an **urban neighbourhood, Serangoon.** o The students have **very limited resources**, and pursuing the study in **a larger area will not be feasible**. **AO1**	**4**
1 (b)	**With reference to Resources 1A and 1B, explain the strengths and weaknesses of the students' choice of sites and neighbourhood.** *Award 1 mark for each strength or weakness.* *Award a maximum of 1 additional mark for further development of each suggestion, where applicable.* *Award a maximum of 3 marks for explanation of <u>either</u> strengths <u>or</u> weaknesses only.* Possible responses include: Strengths:	**6**

Question	Answer	Marks
	• As shown in Resource 1A, the three chosen sites in Serangoon are **located near to one another within walking distance**. o It would be **convenient** for the students to meet and reiterate their data collection methods before their fieldwork and to triangulate the data collected after collecting their data. • As shown in Resource 1A, the three chosen sites are **located near to the town council, at a community hub and near to a bus interchange** which are **key nodes in the neighbourhood**. Weaknesses: • As shown in Resource 1B, there are **textile recycling bins located** at these three chosen sites. When the students conduct their survey at these three chosen sites, they are **introducing sampling bias in their data collection** since they would be approaching youth who have used these bins to be their survey respondents. o This could skew results as the sample might not represent the full spectrum of youth attitudes towards textile recycling. • Focusing only on Serangoon **may not be representative of Singapore's youth population**. **AO2**	
1 (c)	**Explain the strengths and limitations of the sampling method used by the students.** *Award 1 mark for each strength or limitation.* *Award a maximum of 1 additional mark for further development of each suggestion, where applicable.* *Award a maximum of 2 marks for explanation of <u>either</u> strengths <u>or</u> limitations only.* Possible responses include: Strengths: • Convenience sampling is **easy to implement** and allows for quick data collection, given the students' time and resource constraints. o It is practical and easy to **meet their target sample size** of 30 respondents by **approaching people who are readily available**. Limitations: • **Some population groups may be underrepresented or excluded entirely**. For example, it was not	**4**

Question	Answer	Marks
	mentioned that respondents form both genders would be included. o The participants should consist of **both male and female** respondents as the **perceptions, attitudes and actions towards textile recycling and upcycling may vary**. *[Note: If a candidate has included **sampling bias** in his / her response in 1(c), he / she would not be credited for this point if it is repeated in 1 (d).]* **AO1**	
1 (d)	**Using Resource 2, sketch a graph to present the data and justify your choice.** *Award 1 mark each for title, axis, legend, up to a total of 2 marks.* *Award 1 mark for accuracy.* *Award up to 2 marks for justification.* Possible responses: Justification • A **comparative bar graph or compound bar graph** helps to present multiple data sets in 1 graph sharing the same axes. • This allows **easy comparison of data sets** across the 3 sites. **AO2**	**5**
1 (e)	**With reference to Resource 3, explain the strengths of the survey form used.** *Award 1 mark for each explanation of strength, to a maximum of 6 marks.* *Award a maximum of 1 additional mark for further development of each explanation.* *Note: Only about the survey questions, not about the time or sampling.* Possible responses include: **Strengths** • It has both open-ended and close-ended questions. This would allow them to collect both **quantitative and qualitative data**. The open-ended question allows participants to efficiently answer the survey with the given choices.	**6**

Question	Answer	Marks
	o The students can also quickly tabulate the quantitative data based on the pre-defined choices. • The design includes background information by asking for **age and residency** which ensures that the respondents **fall within target group of youth residing in Serangoon**. • Having the background information of **gender and education level** helps **categorise respondents** and understand the **demographic background** of the sample. This allows for **deeper analysis and better interpretation** of attitudes, behaviours and knowledge levels between genders and different education groups. • The survey includes **valid questions** which are well-crafted with a range of responses and can be used to test the hypothesis. o The survey also includes questions, e.g. Q1 and Q2, to investigate the respondents' awareness of environmental impact of textiles industry and textile waste. **AO2**	
1 (f)	**Suggest how the students could improve the accuracy and reliability of their data in Resource 4.** *Award 1 mark for each suggestion of accuracy or reliability, to a maximum of 5 marks.* *Award 1 mark for further development of each suggestion, where applicable. Award a maximum of 3 marks if the response only addresses accuracy or reliability of data.* Possible responses include: To improve **accuracy** (*how close the data is to the truth or what it aims to measure*): • **Use stratified or systematic sampling instead of convenience sampling**. o This **reduces sampling bias** compared to just asking whoever seems convenient and ensures results better reflect the youth population in Serangoon. • **Include a more diverse range of sites** rather than in Serangoon only. Captures a broader and more accurate picture of youth behaviour in different areas. To improve **reliability** (*how consistent and repeatable the data collection process is*):	**5**

Question	Answer	Marks
	• **Repeat the data collection during different times of the day and on multiple days**, including both weekdays and weekends. • Ensure the students have a discussion / training session to ensure they have **consistent interpretation and administration of questions**. **AO2**	
1 (g)	**With reference to all the resources, evaluate the validity of their conclusion.** Possible Approaches: Candidates could evaluate the validity of their conclusion by **reviewing the hypothesis and choice of research area / locations** shown in Resources 1A and 1B and **the methods used to collect data and the data collected** shown in Resources 2, 3, 4 and 5. The study had a **clear focus and hypothesis** and directly measured both **awareness** of environmental impact and **participation** in textile recycling and upcycling. For data collection methods, candidates **could review each of the methods (observation, questionnaire survey and in-depth interviews) in turn and / or review both collectively**, including the **order** in which they were carried out, to assess the impact on **accuracy and reliability of data collected**. Evaluation could also look into the **sample size of the questionnaire survey** and whether the sample selected is **representative** of the youth population in Serangoon (besides considering gender, age and education level segments, ethnicity could be considered too), **strengths and weaknesses** of the different methods, etc. Candidates could consider the **design of the questionnaire survey,** such as its limitations (did not consider variations across ethnic groups). **Interviews** helped to better **reveal personal motivations, values and peer influence**, which are hard to measure in tick-box survey questions. Levels marked using modified Generic Level Descriptors for H2 fieldwork evaluative questions. **AO3**	**10**

Section B

Cluster 3: Sustainable Future and Climate Change

Question	Answer	Marks
2 (a)	**Define sustainable urban development.** *Award a maximum of 2 marks for the definition of sustainable urban development.* Possible responses include: • Sustainable urban development is development that **meets the needs of present generation** • **without compromising on the ability of future generations to meet their needs** in the **urban context**. **AO1**	**2**
2 (b)	**Describe how the indicator of Sustainable Cities Index can be used to measure sustainable urban development.** *Award 1 mark for each description of how the indicator is used to measure SUD.* *Award a 1 additional mark for further development of each description, where applicable.* Possible Responses include: • The Sustainable Cities Index **measures cities' performance across three pillars of planet, people and profit (and a newly introduced progress pillar)** that mirrors the **triple bottom line of sustainable development.** • It is a **composite indicator** that allows cities to measure its sustainability **more holistically** across the different dimensions. • The **People** pillar measures **social performance including quality of life** such as inclusivity, safety, providing opportunities for work and ease of access to transport and healthcare. • The **Profit** pillar assesses **business environment and economic performance**, considering aspects such as GDP per capita, ease of doing business, affordable housing, and other economic measures. • The **Planet** pillar captures **environmental factors like energy usage and emissions**. The index investigates climate-related threats to cities as well as policies needed to bolster a city's environmental goals. **AO1**	**5**

2 (c)	**Compare the ranking of the cities shown in Resource 6.** *Award 1 mark for each comparison of the ranking, to a maximum of 6 marks.* *Award 1 additional mark for further development of each comparison, where applicable.* Possible responses include: • **Western European cities outperformed Northern Europe and the rest of the world.** o The **top overall sustainable cities** of **Amsterdam and Rotterdam** are both in the Netherlands, i.e. Western European cities. o Out of the top 3 positions for Overall and the 4 pillars which yielded a total of 15 positions, **Western European cities swept almost half of them.** • **Northern European cities fared better in the Planet pillar** while **Western European cities did better in the People pillar**. o **Oslo and Copenhagen** are ranked top 2 in the **Planet** pillar, while **Paris and Vienna** swept the top 2 positions for the **People** pillar. • **North American cities** fared better in the **Profit** pillar while **Asian and African cities** clinched the top 3 for the **Progress** pillar. **AO2**	**6**
2 (d)	**Suggest reasons for the ranking described in (c).** *Award 1 mark for each reason of the ranking, to a maximum of 5 marks..* *Award 1 additional mark for further development of each reason, where applicable.* Possible Responses include: • **Environmental sustainability is a firmly established strength among European cities**, **particularly Northern European cities**, who are leading the way in this most fundamental measure of sustainability. • This can be attributed to their **governance, innovation, human capital and climate action efforts**, which allows it to boost its score in the Planet pillar. • **Western European cities** are generally **global leaders in social sustainability**, due to strong urban planning, social safety nets, policies and inclusive infrastructure. • **North American cities**, especially American cities, are generally **global leaders in economic sustainability**	**5**

	due to their **economic strength and resilience,** strong GDP, global business connectivity and skilled workforce. • **Asian and African cities** score highly for the **Progress pillar** because they are starting from lower bases and advancing rapidly in areas like **technology, infrastructure and human development**. **AO1**	
2 (e)	**With reference to Resources 7 and 8, explain the possible challenges in progressing towards sustainable urban development in Kinshasa.** *Award 1 mark for each explanation of the possible challenges. Award 1 additional mark for further development of each explanation, where applicable.* Possible Responses include: • From Resource 7, it is evident that Kinshasa faces rapid **urbanisation and population growth** which poses challenges to sustainable urban development. o With DRC's national population increasing from 71.4 m in 2013 to 105.8 m in 2023, it is likely that Kinshasa is a fast-growing city mirroring its national demographic trend. o Without careful planning, this growth can lead to unregulated urban sprawl, slums and strain on public services, undermining sustainability. • With **a small proportion of people employed in the manufacturing sector** (9.8% in 2023) and **a large proportion in the agriculture sector** (55% in 2023), the country is largely dependent on primary industries, such as agriculture and extractive industries. o Over-reliance on primary industries makes the economy vulnerable to price shocks, e.g. fluctuations in global prices of commodities. o Resource 6 shows street hawkers selling food next to the road. A significant proportion of the urban dwellers is working in the informal economy with low job security and low wages. • With **low GDP per person** at only US$ 664 in 2023, the **affordability of housing or services** would be an issue to many urban dwellers. **AO2**	**7**
2 (f)	**Explain how the vulnerability of different groups of people may result in varied impacts of climate change.**	**5**

	Award 1 mark for each explanation of differences, to a maximum of 5 marks. *Award 1 additional mark for further development of each explanation, where applicable.* Possible Responses include: • Differences in vulnerability among various groups of people mean that climate change does not affect everyone equally. **Factors such as age, income, gender and access to resources can significantly influence** how people experience and cope with the impacts of climate change. • **Income and socioeconomic status: Higher-income groups** could afford **adaptation measures** like air conditioning, flood insurance or relocation to cope with impact of climate change. o Poorer communities often live in high-risk areas and lack the financial resources to recover from hazards such as tropical cyclones and floods. • **Gender: Women and girls often bear the brunt of climate change in CLLDs** due to **gender roles and inequalities**. In many communities, women are responsible for gathering water and food. • **Age and health**: **Children, the elderly and people with pre-existing health conditions** are more vulnerable to extreme heat, poor air quality and disease outbreaks due to their lower immunity. **AO1**	

Section C

Cluster 3: Sustainable Future and Climate Change

Question	Answer	Marks
3	**Creating cities that are liveable is more achievable than making them truly sustainable.'** **Evaluate this statement**. Possible Approaches: Candidates could approach the question by making a judgement about **whether liveability goals** such as safety, comfort, transport access and creation of green spaces **are easier to achieve than sustainability goals** such as reducing emissions, limiting ecological footprints and achieving intergenerational equity. These **liveability improvements** can yield **short-term**	**20**

	gains while **sustainability** requires **long-term planning**, trade-offs and managing political-economic challenges. Candidates could evaluate how **improvements in liveability or short-term solutions can sometimes conflict with long-term sustainability, i.e. tension**. Candidates could analyse the context in which **developed cities can afford to pursue both goals more readily than developing cities**. Alternatively, the **complementarity of these strategies** can also be examined. Additionally, candidates may apply various criteria to evaluate the relative achievability of liveability and sustainability goals including **stakeholder support, political and social feasibility and measurability**. *Levels marked using H2 generic level descriptors for 20m H2 essays.* **AO3**	

Question	Answer	Marks
4	**'The countries least responsible for climate change often suffer its harshest consequences and face challenges in implementing effective responses.'** **Evaluate this statement.** Possible Approaches: Candidates could approach the question by assessing the **disproportionate disparity between responsibility and impact**. This could entail making a judgement on how **countries with low historical greenhouse gas emissions or low per capita emissions often experience the most severe climate impacts**. Candidates could analyse the contexts of countries facing sea level rise, drought and desertification and increased hazard frequency and intensity, e.g. tropical cyclones. Candidates could present various **counterarguments** to the statement by examining how **countries least responsible** for climate change have demonstrated **strong climate leadership**, as well as how **some developed nations, though responsible,** also suffer **harsh impacts**. The evaluation could include considerations of countries that have **better response capacity** and the role of **equity and justice** at international scales including COP discussions and the Paris Agreement. *Levels marked using H2 generic level descriptors for 20m H2 essays.*	**20**

	AO3	